# REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

**Generated:** 2026-09-14 13:24:49
**Reporting Period:** All Recorded Traffic

## 1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

| Metric | Value |
|---|---|
| Total Network Flows | 11613 |
| Normal Flows | 11055 |
| Detected Attacks | 558 |
| Attack Rate | 4.80% |

## 2. Machine Learning Model

| Parameter | Value |
|---|---|
| Algorithm | Random Forest Classifier |
| Training Features | duration, protocol_type, src_bytes, dst_bytes, count |
| Classification | Normal / Attack |
| Optimized Threshold | 0.50 |
| Validation Accuracy | 0.76% |
| Attack Precision | 0.96 |
| Attack Recall | 0.60 |
| Attack F1-Score | 0.74 |

## 3. Network Protocol Distribution

| Protocol | Flow Count |
|---|---|
| TCP | 3923 |
| UDP | 3194 |
| HTTPS | 2819 |
| DNS | 923 |
| QUIC_OR_UDP443 | 671 |
| HTTP | 50 |
| IPSEC | 16 |
| DHCP | 13 |
| OTHER | 2 |
| ICMP | 1 |
| NTP | 1 |

# 4. Detected Attack History

| Time | Source | Destination | Protocol | Threat Vector | Severity | Confidence |
|---|---|---|---|---|---|---|
| 2026-09-06 00:48:54 | 192.168.1.7:57688 | 8.8.8.8:443 | HTTPS | ANOMALOUS FLOW | HIGH | 100.0% |
| 2026-09-06 00:48:53 | 192.168.1.7:64056 | 8.8.8.8:443 | QUIC_OR_UDP_443 | ANOMALOUS FLOW | HIGH | 100.0% |
| 2026-09-06 00:48:52 | 104.16.9.34:443 | 192.168.1.7:56577 | HTTPS | ANOMALOUS FLOW | HIGH | 76.0% |
| 2026-09-06 00:48:52 | 104.16.9.34:443 | 192.168.1.7:56575 | HTTPS | ANOMALOUS FLOW | HIGH | 57.39% |
| 2026-09-06 00:48:52 | 104.16.9.34:443 | 192.168.1.7:56576 | HTTPS | ANOMALOUS FLOW | HIGH | 68.0% |
| 2026-08-29 20:33:34 | 10.94.204.202:60791 | 10.94.204.69:53 | DNS | ANOMALOUS FLOW | HIGH | 92.0% |
| 2026-08-29 20:33:34 | 10.94.204.202:51449 | 10.94.204.69:53 | DNS | ANOMALOUS FLOW | HIGH | 88.0% |
| 2026-08-29 20:33:34 | 10.94.204.202:52013 | 10.94.204.69:53 | DNS | ANOMALOUS FLOW | HIGH | 74.0% |
| 2026-08-29 20:33:34 | 10.94.204.202:54332 | 10.94.204.69:53 | DNS | ANOMALOUS FLOW | HIGH | 70.0% |
| 2026-08-29 20:33:33 | 142.251.151.119:443 | 10.94.204.202:55747 | QUIC_OR_UDP_443 | ANOMALOUS FLOW | HIGH | 73.59% |
| 2026-08-29 19:53:23 | 10.94.204.202:65368 | 172.217.118.4:443 | HTTPS | ANOMALOUS FLOW | HIGH | 77.36% |
| 2026-08-29 19:53:04 | 10.94.204.202:55082 | 140.82.113.21:443 | HTTPS | ANOMALOUS FLOW | HIGH | 50.34% |
| 2026-08-29 19:48:54 | 10.94.204.202:65368 | 172.217.118.4:443 | HTTPS | ANOMALOUS FLOW | HIGH | 73.89% |
| 2026-08-29 19:40:42 | 10.94.204.69:53 | 10.94.204.202:61152 | DNS | ANOMALOUS FLOW | HIGH | 84.67% |
| 2026-08-29 19:39:49 | 10.94.204.69:53 | 10.94.204.202:53580 | DNS | ANOMALOUS FLOW | HIGH | 53.86% |
| 2026-08-29 19:38:55 | 10.94.204.202:62659 | 172.64.155.209:443 | HTTPS | ANOMALOUS FLOW | HIGH | 53.38% |
| 2026-08-29 18:17:22 | 10.94.204.202:60479 | 172.64.155.209:443 | HTTPS | ANOMALOUS FLOW | HIGH | 54.24% |
| 2026-08-26 22:10:09 | 192.168.1.1:53 | 192.168.1.5:52851 | DNS | ANOMALOUS FLOW | HIGH | 93.11% |
| 2026-08-26 22:10:09 | 192.168.1.1:53 | 192.168.1.5:60211 | DNS | ANOMALOUS FLOW | HIGH | 93.11% |
| 2026-08-26 22:10:09 | 192.168.1.1:53 | 192.168.1.5:63849 | DNS | ANOMALOUS FLOW | HIGH | 85.45% |

# 5. Security Recommendations

**1.** Investigate repeated attack predictions from the same source address.

**2.** Monitor network flows with unusually high traffic volume or packet counts.

**3.** Review suspicious TCP and UDP connections identified by the ML classifier.

**4.** Use the attack confidence score together with network context when validating alerts.

**5.** Continue collecting representative network traffic to improve the IDS model.

**6.** Periodically retrain and evaluate the model using updated datasets.

# 6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.